Lab 10 — Real-Time Data Ingestion into Redshift using Kinesis Firehose

Lab Summary

In this lab, you will build a real-time ingestion pipeline where:

Kinesis Data Stream → Kinesis Firehose → Lambda (data enrichment) → Redshift

Streaming clickstream data will be enriched using Lambda and ingested into Redshift Serverless for real-time reporting.

Prerequisites

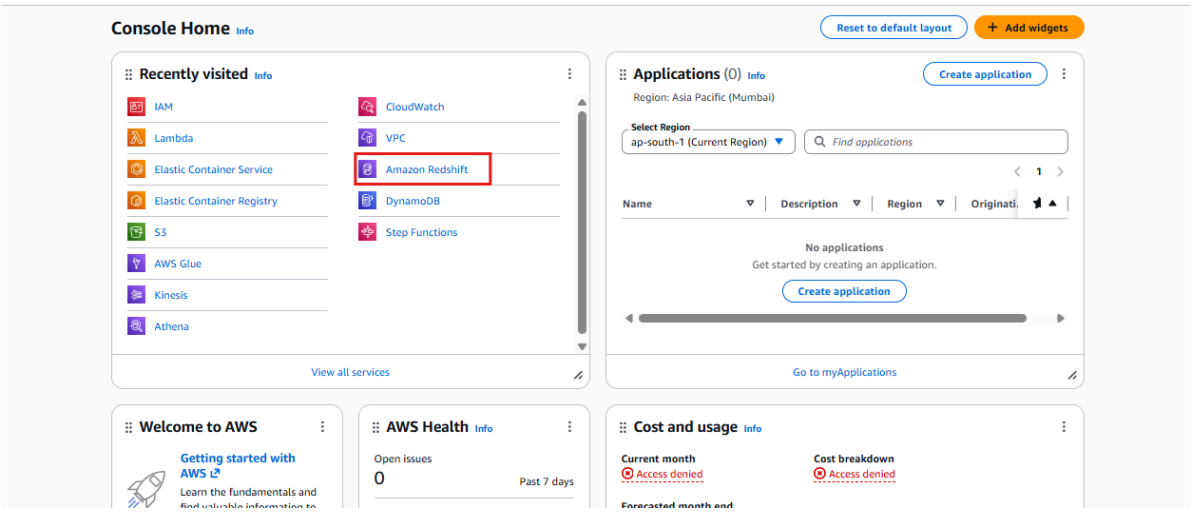
- AWS Account
- Redshift Serverless workgroup created
- S3 bucket created (for Firehose intermediate storage)
- IAM role with:
  - Kinesis access
  - Lambda access
  - Redshift access
  - S3 access
  - CloudWatch logs access
- events.csv dataset
- Kinesis producer script ready

Steps to Complete Lab

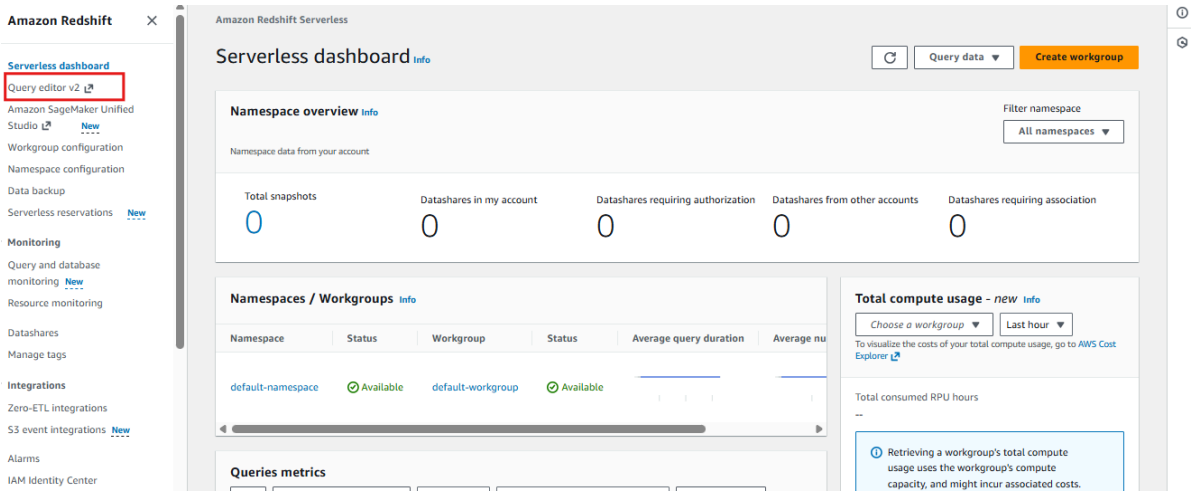
Task 1 — Create Redshift Database & Table

Step 1: Open Redshift Query Editor

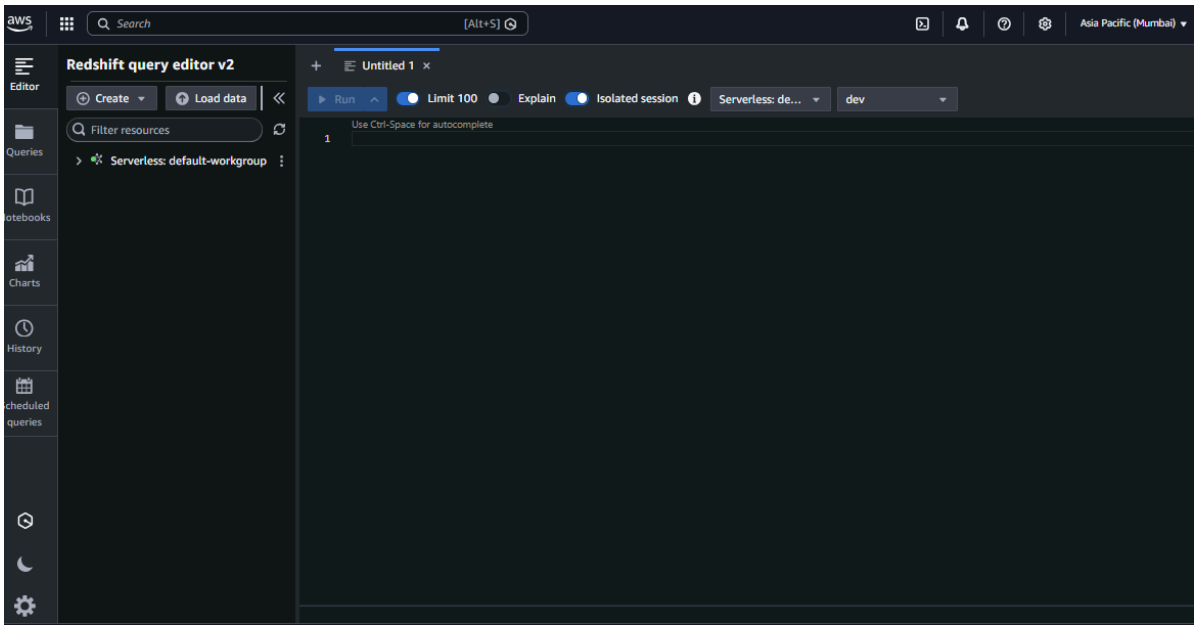
1. Go to AWS Console —→ Search → **Redshift**



2. Click → **Query Editor**



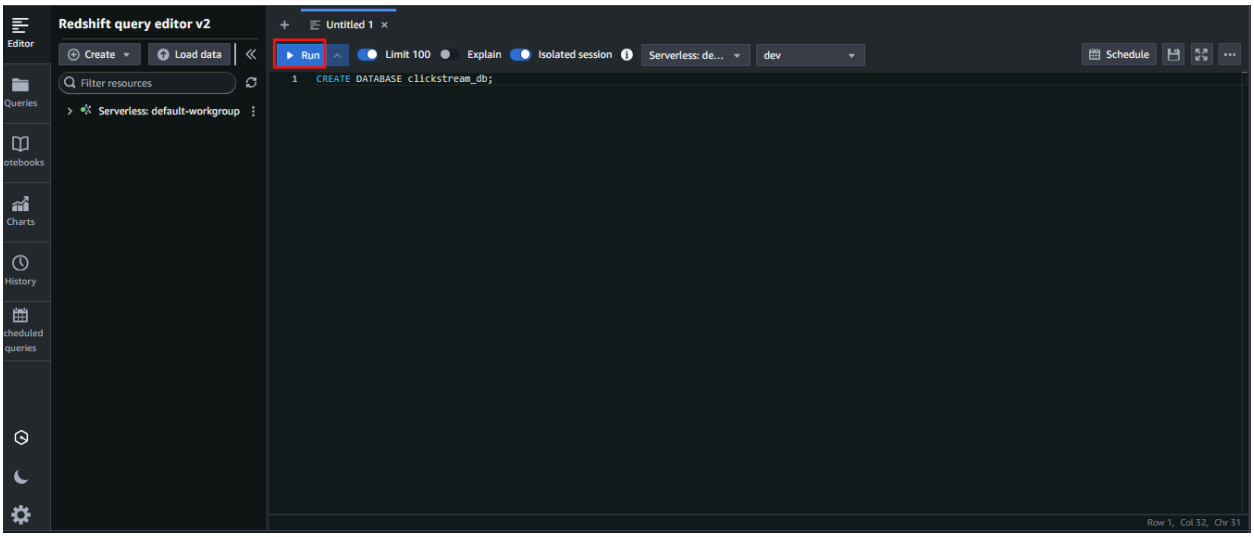
3. Open Query Editor



Step 2: Create Database

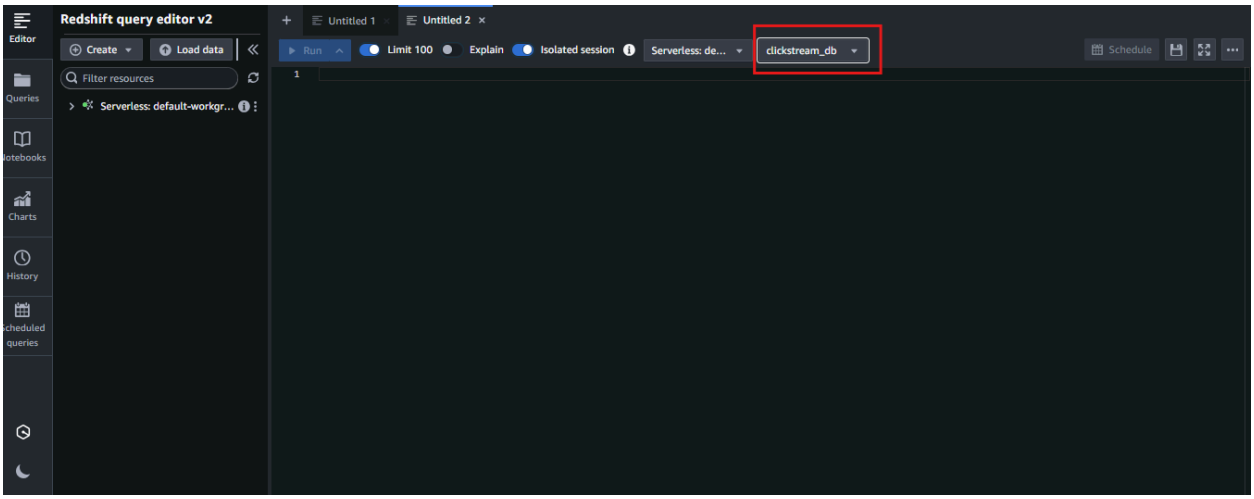
Run:

```
CREATE DATABASE clickstream_db;
```



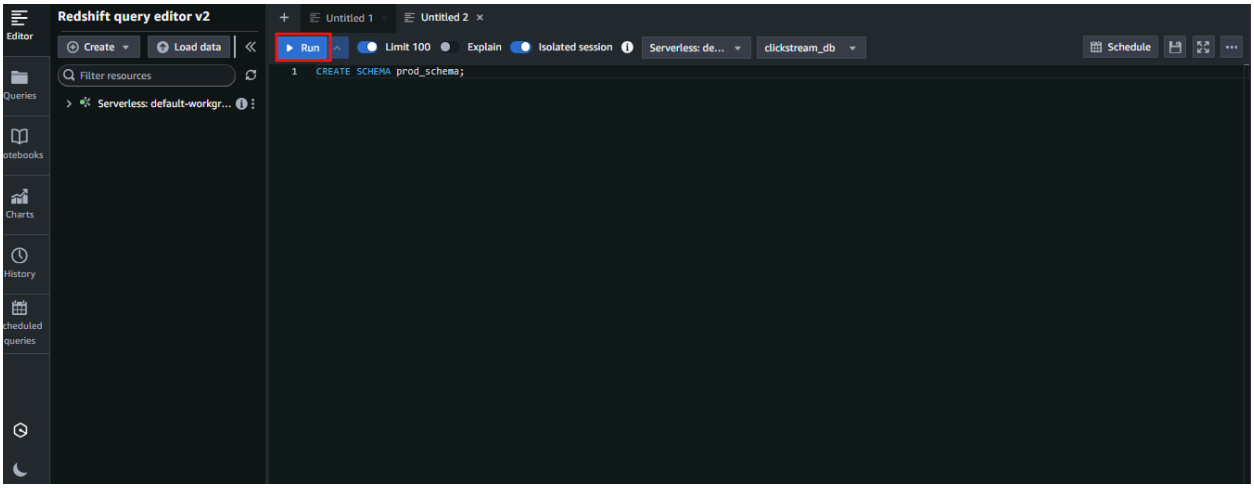
Step 3: Connect to Database

Select database → `clickstream_db`



Step 4: Create Schema

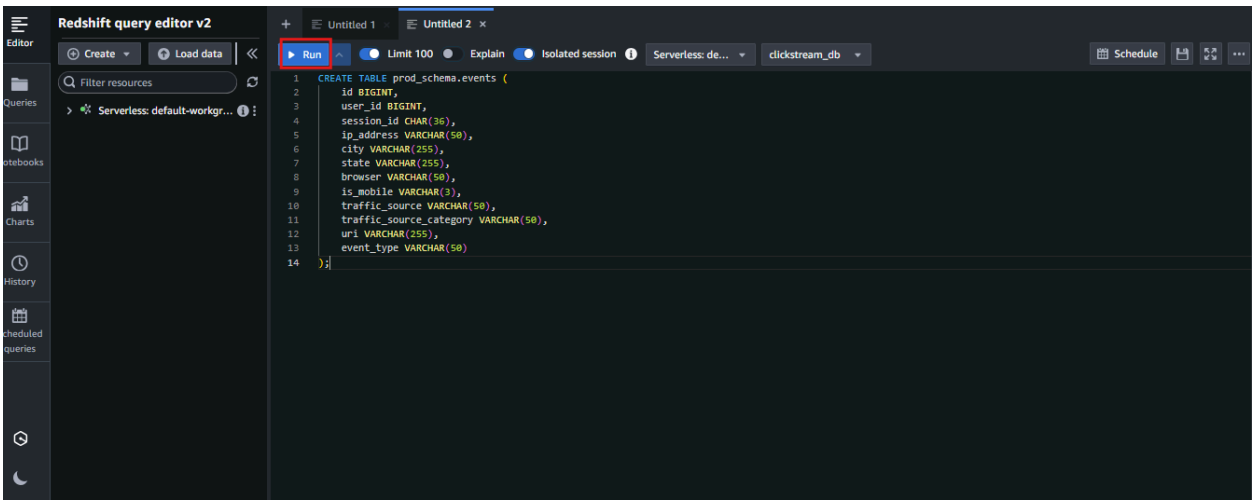
```
CREATE SCHEMA prod_schema;
```



Step 5: Create Events Table

```
CREATE TABLE prod_schema.events (
    id BIGINT,
    user_id BIGINT,
    session_id CHAR(36),
    ip_address VARCHAR(50),
    city VARCHAR(255),
    state VARCHAR(255),
```

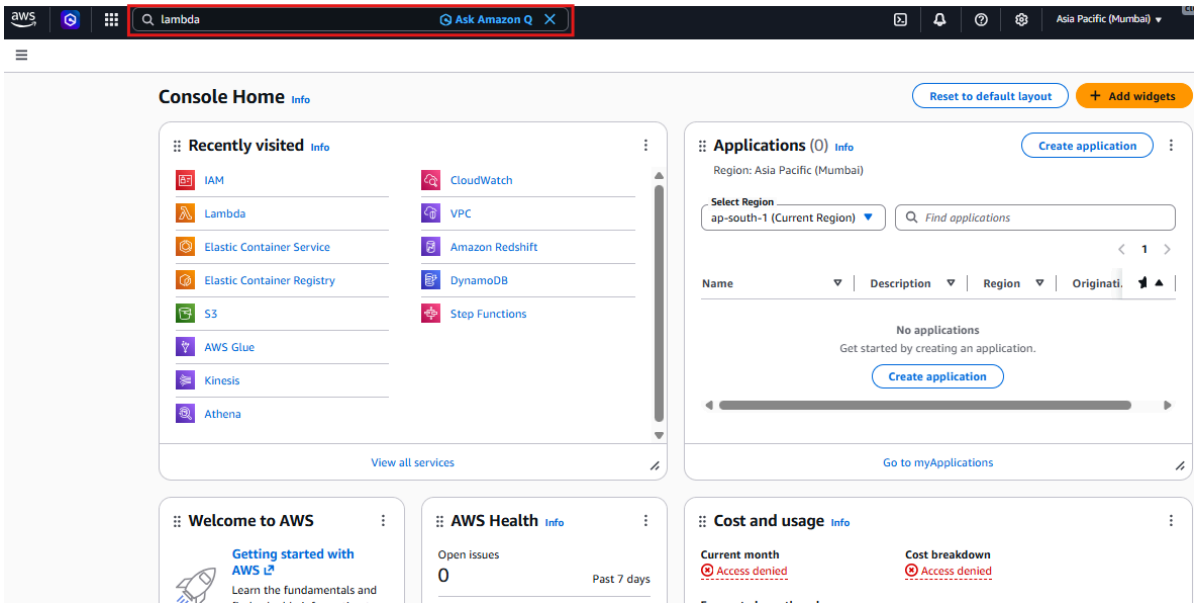
```
    browser VARCHAR(50),
    is_mobile VARCHAR(3),
    traffic_source VARCHAR(50),
    traffic_source_category VARCHAR(50),
    uri VARCHAR(255),
    event_type VARCHAR(50)
);
```



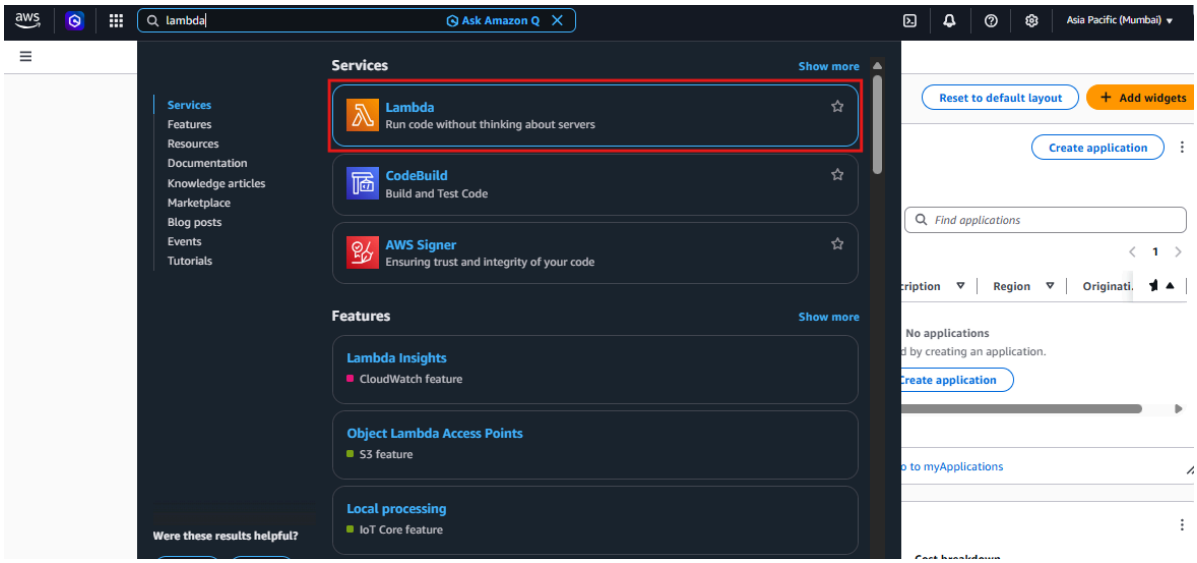
Task 2 — Deploy Lambda Function (Data Enrichment)

Step 1: Create Lambda

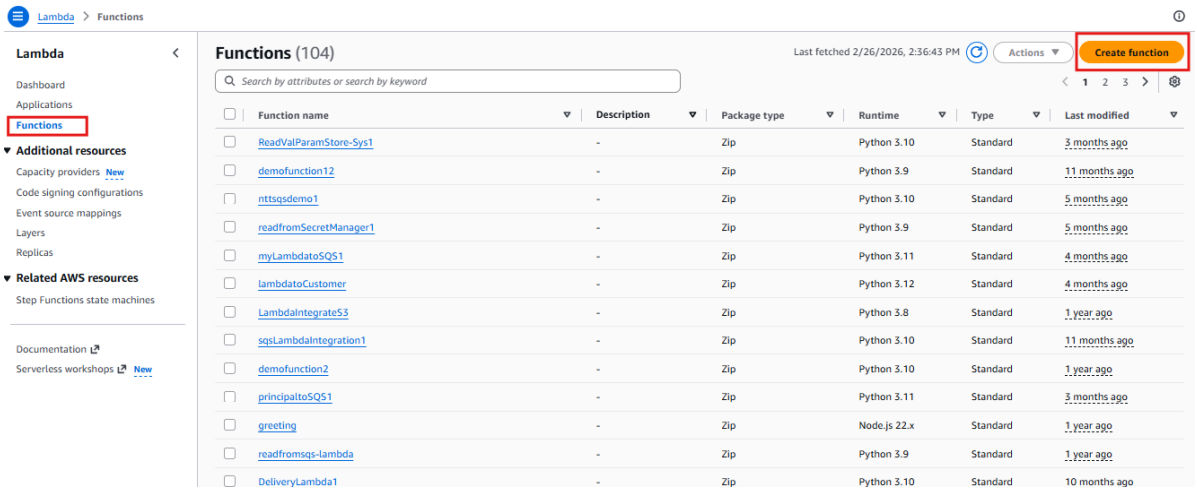
1. Go to AWS Console



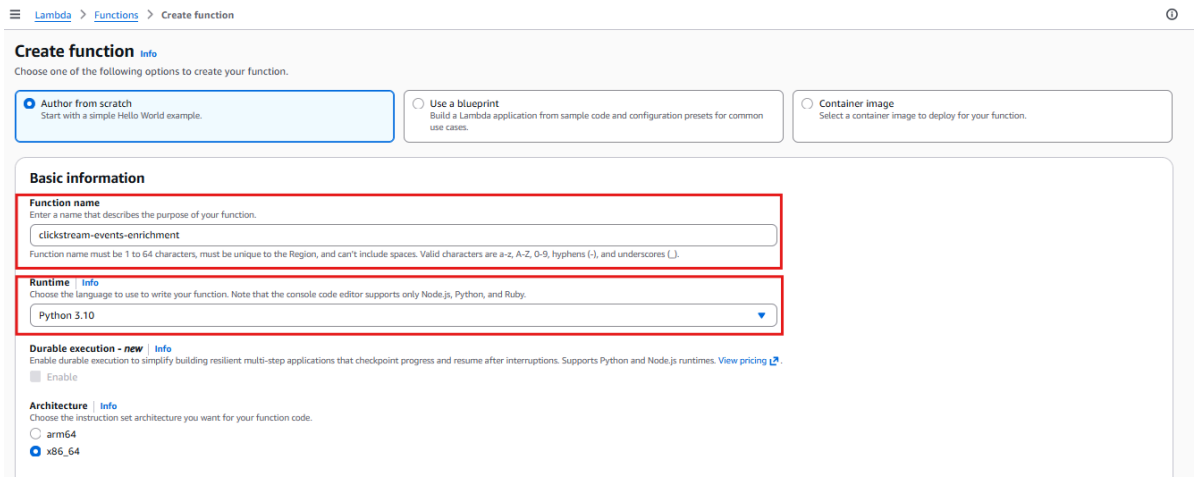
2. Search → **Lambda**



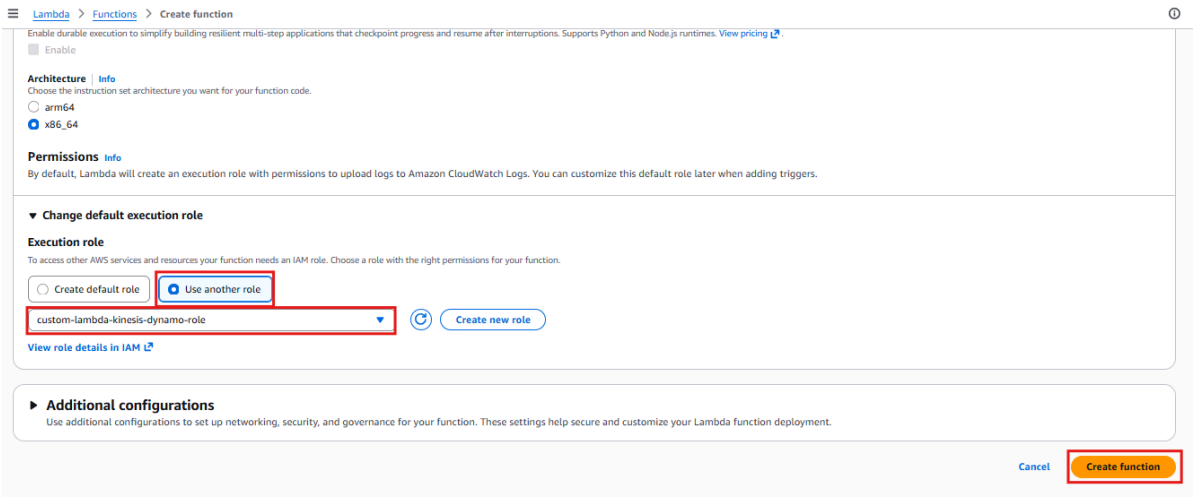
3. Click → Create Function



4. Name → `clickstream-events-enrichment`
5. Runtime → Python 3.10

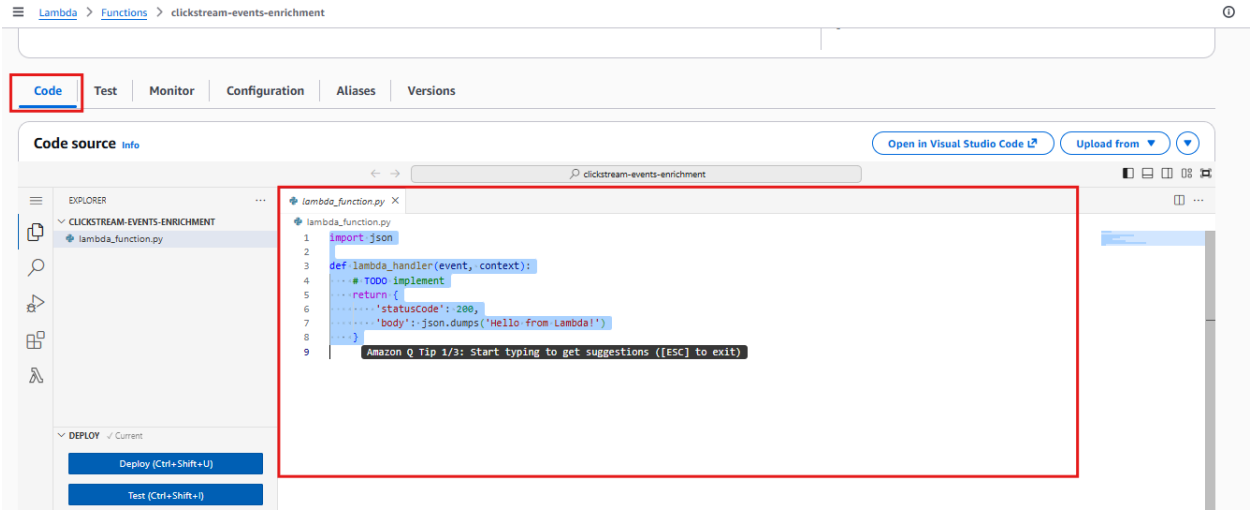


6. Choose an IAM role with:
   - Kinesis access
   - Lambda access
   - Redshift access
   - S3 access
   - CloudWatch logs access
7. Click Create

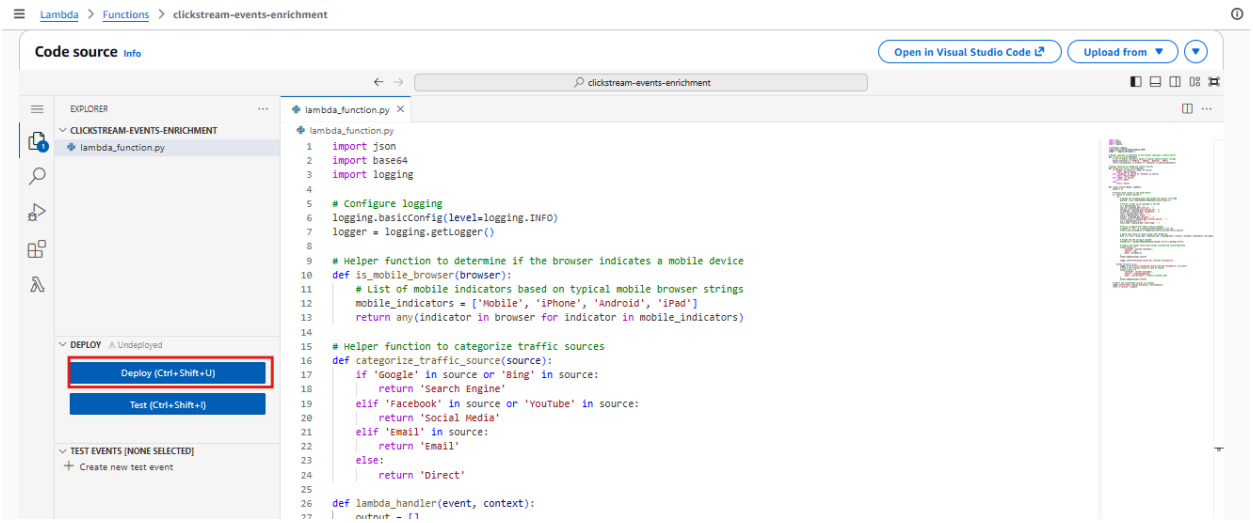


Step 2: Paste Lambda Code

Replace function code —→ lambda.py code

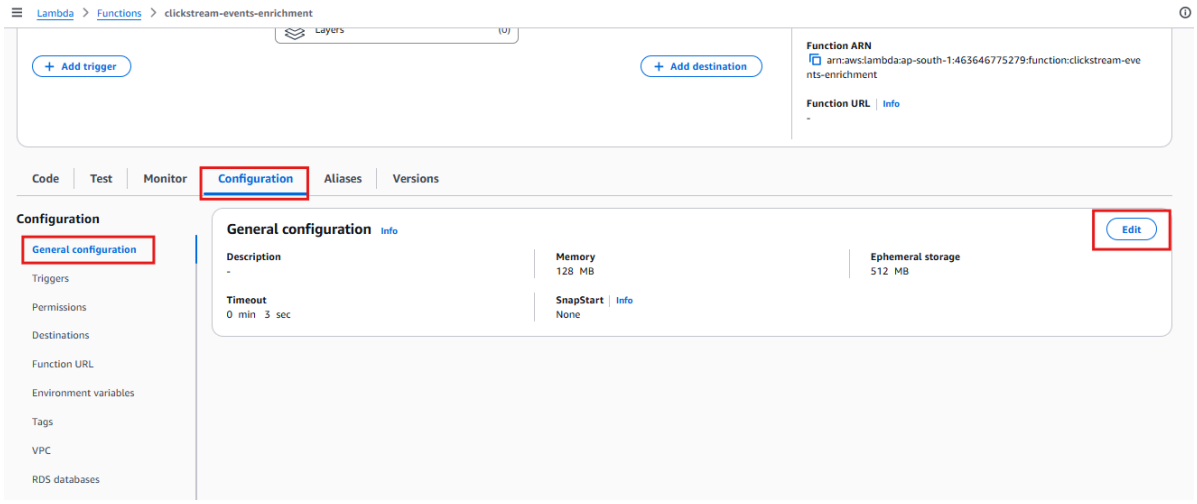


Click **Deploy**.

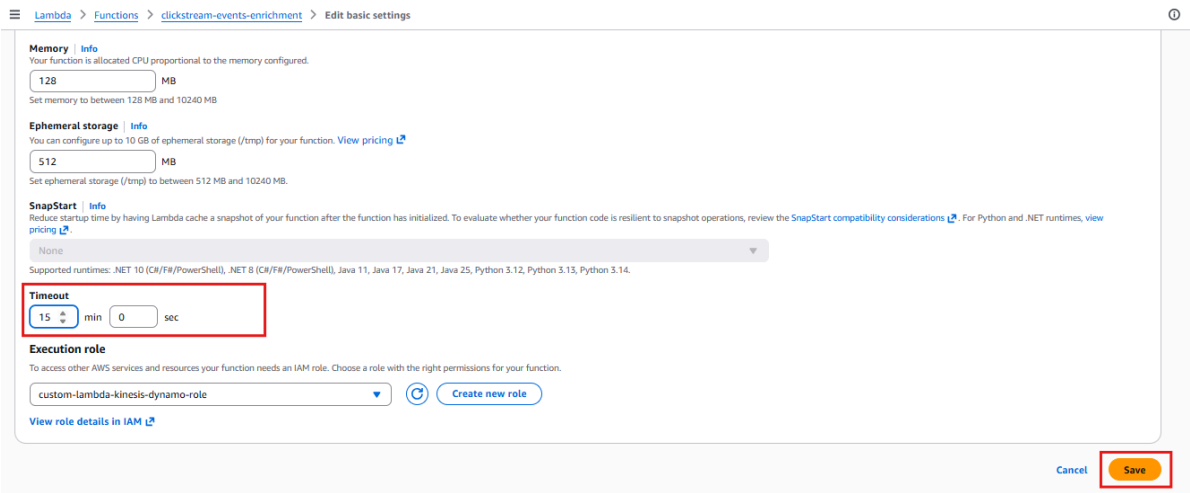


Step 3: Increase Timeout

1. Go to Configuration → General Configuration —→ Click Edit



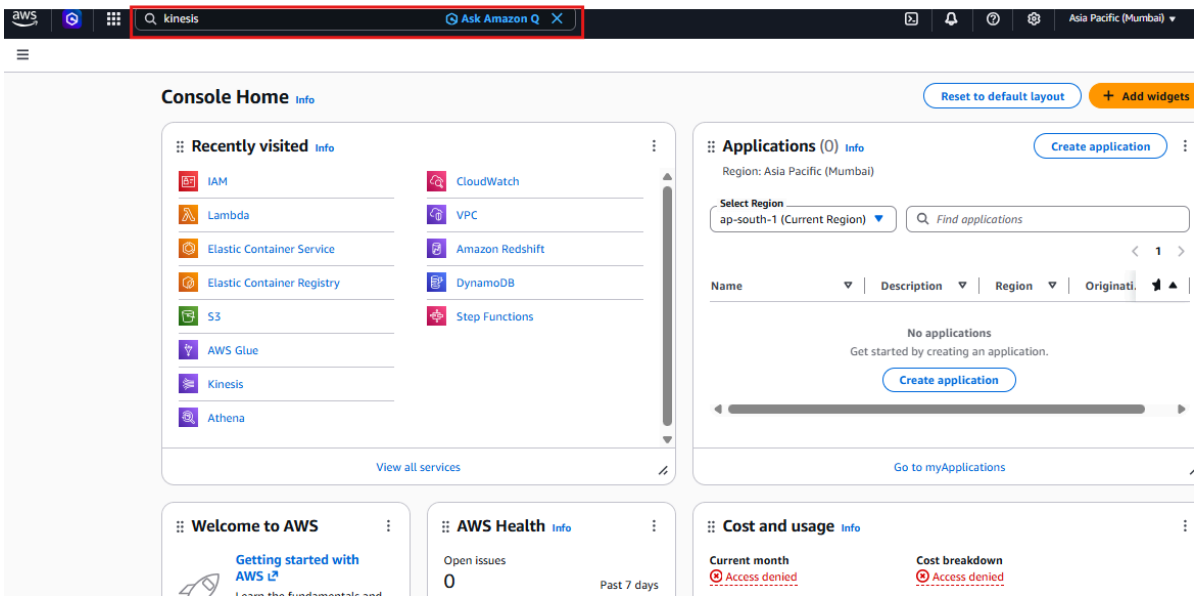
2. Set Timeout → 15 minutes —→ Click Save



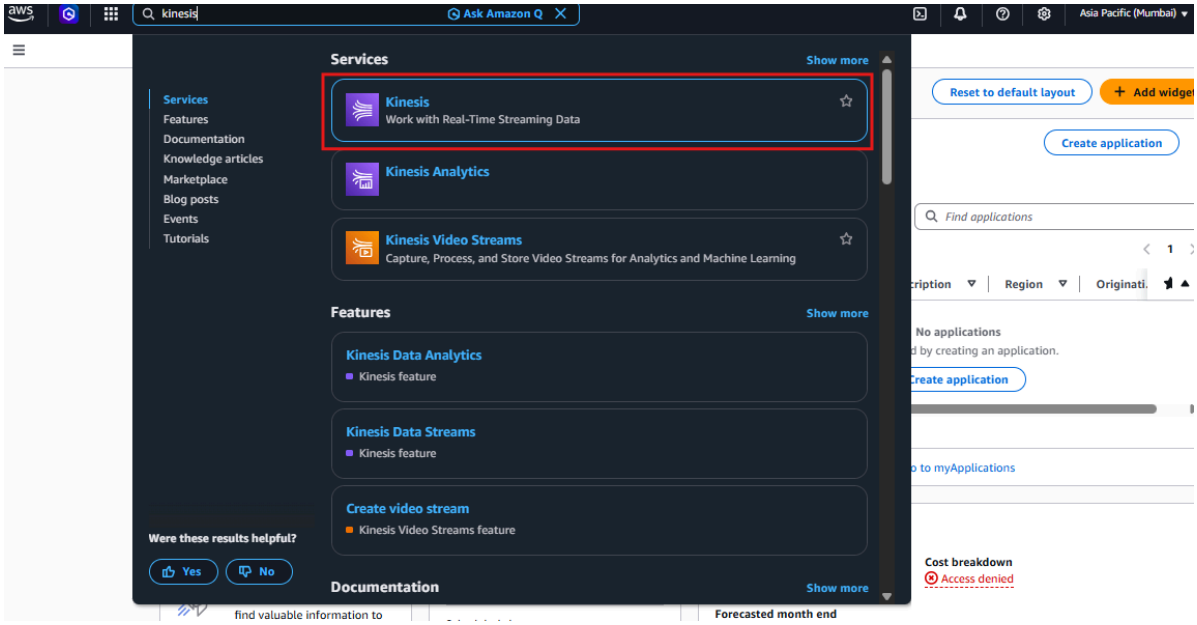
Task 3 — Create Kinesis Data Stream

Step 1: Create Stream

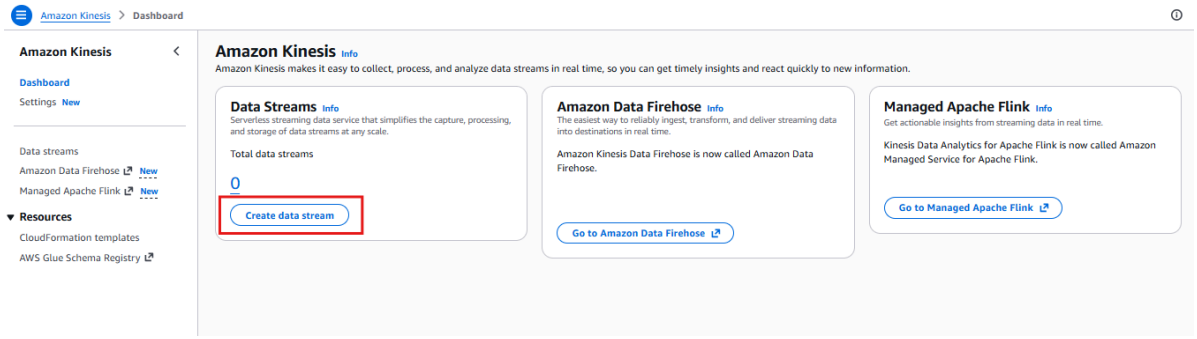
1. Go to AWS Console



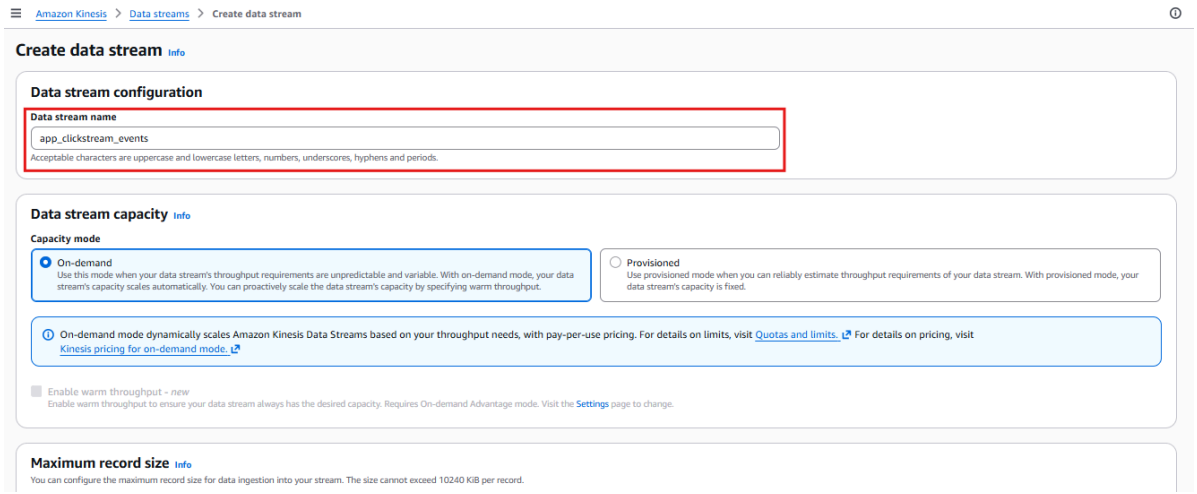
2. Search → **Kinesis**



3. Click → Data Streams —→ Create Data Stream



4. Name → `app-clickstream-events`

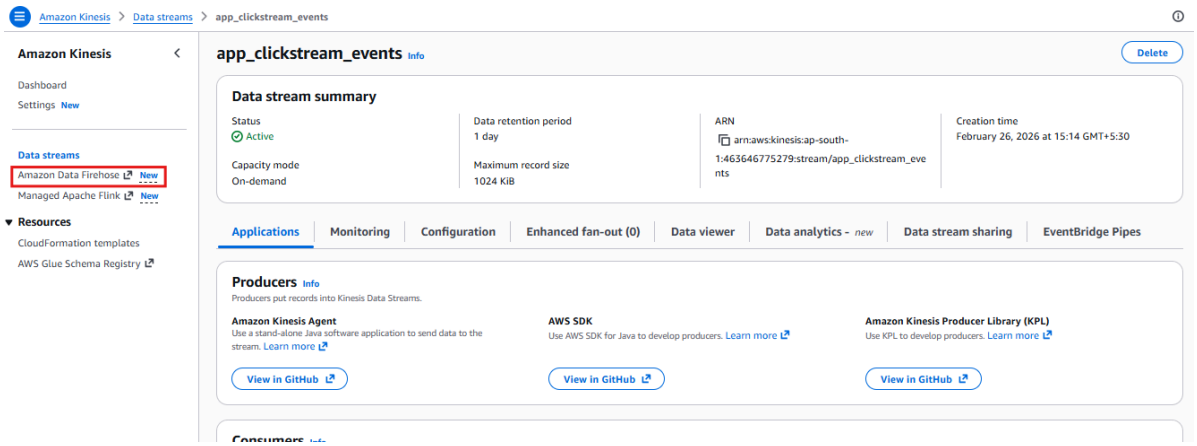


5. Keep default shard settings
6. Click Create data stream

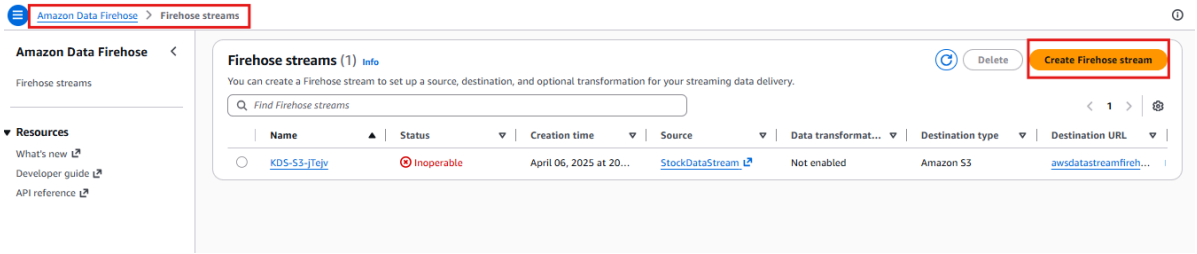
Task 4 — Create Kinesis Firehose Stream

Step 1: Create Firehose

1. In Kinesis Console —→ Click Data Firehose

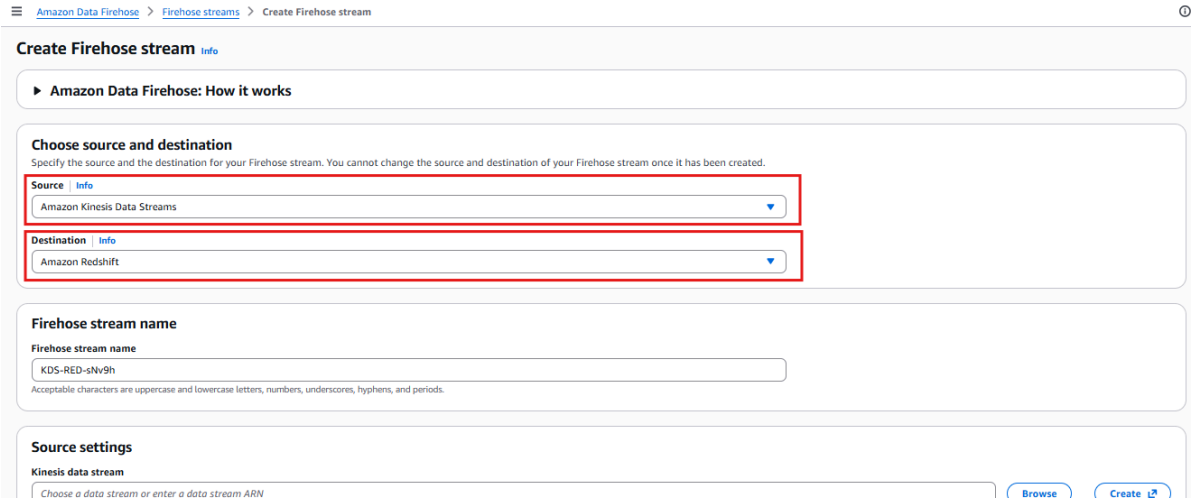


2. Create Firehose

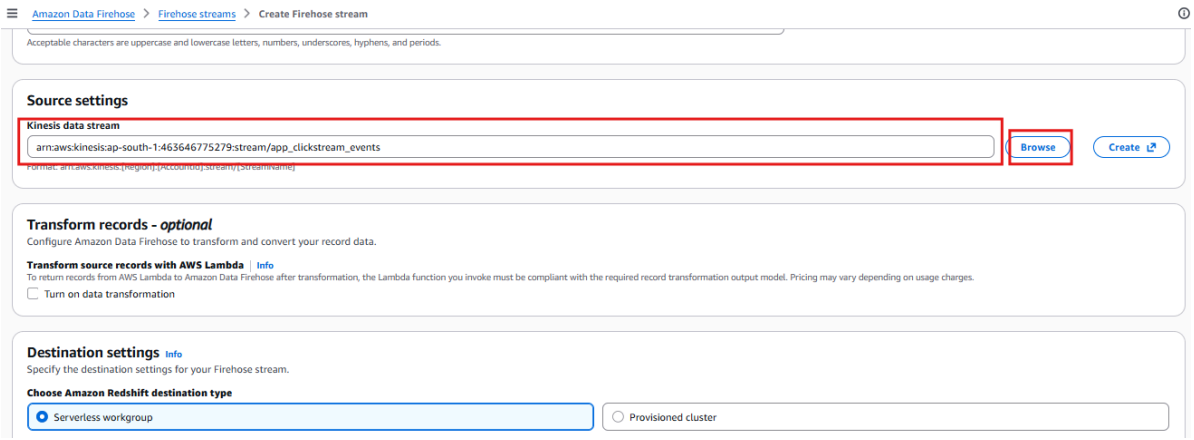


Step 2: Configure Source

- Source → Kinesis Data Stream
- Destination —→ Amazon Redshift

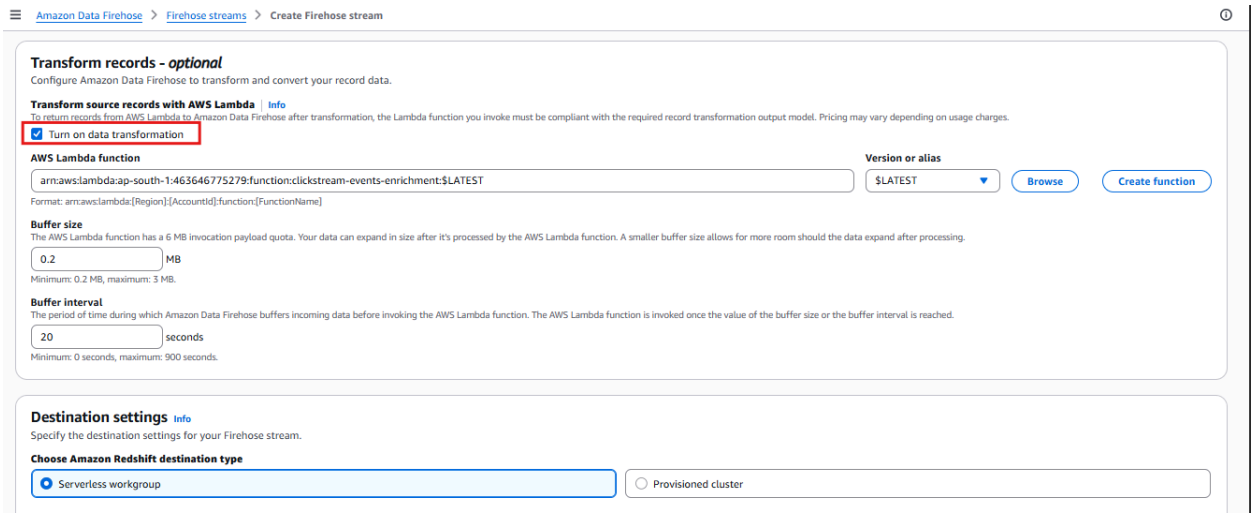


- Source settings —→ Click Browse → Select → `app-clickstream-events`

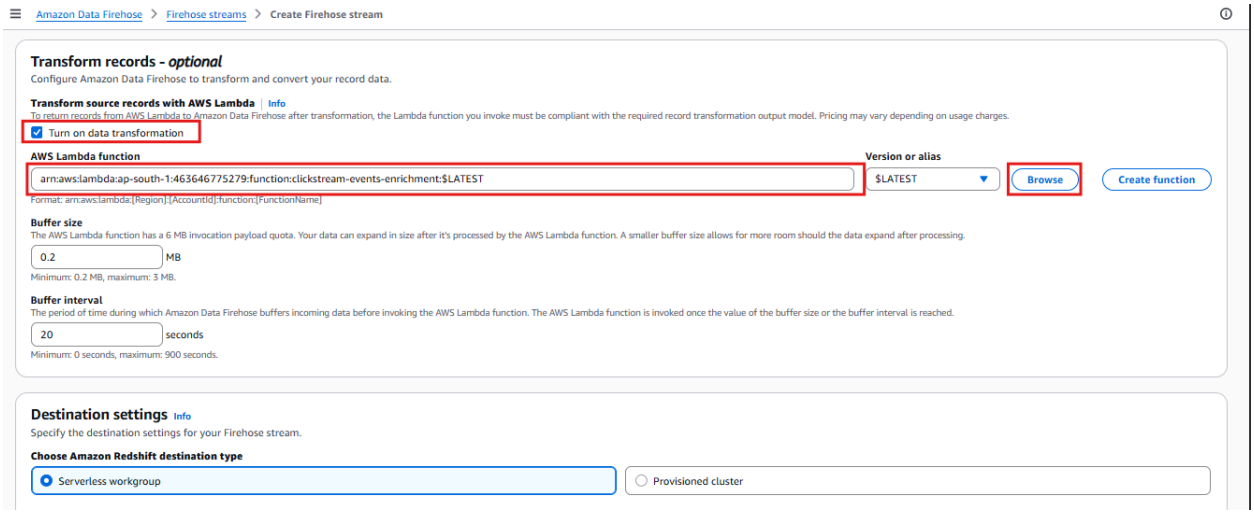


Step 3: Enable Data Transformation

Enable Transform Records

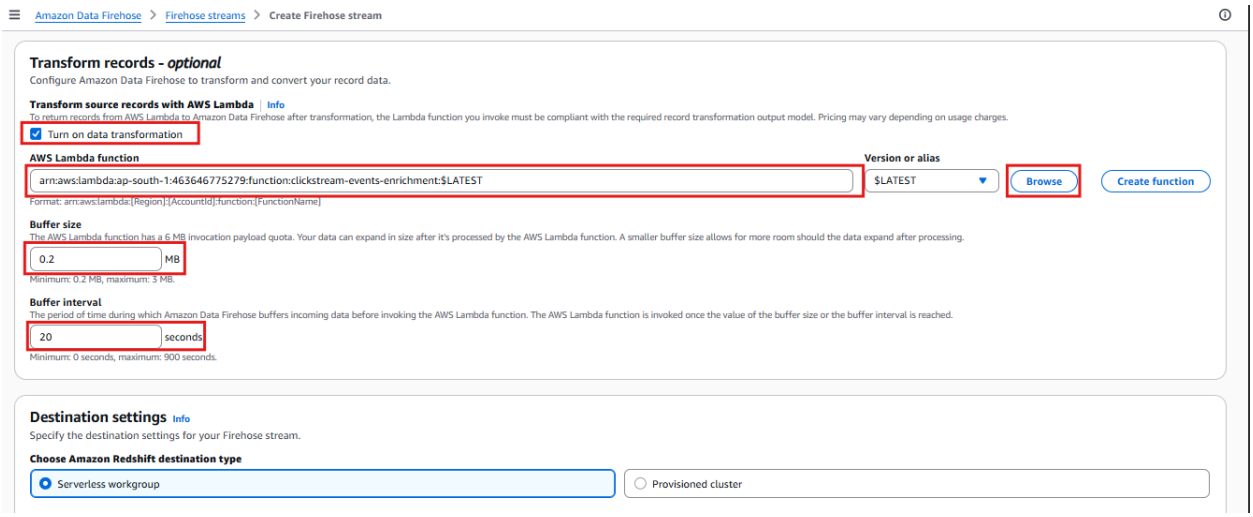


Select Lambda → `clickstream-events-enrichment`



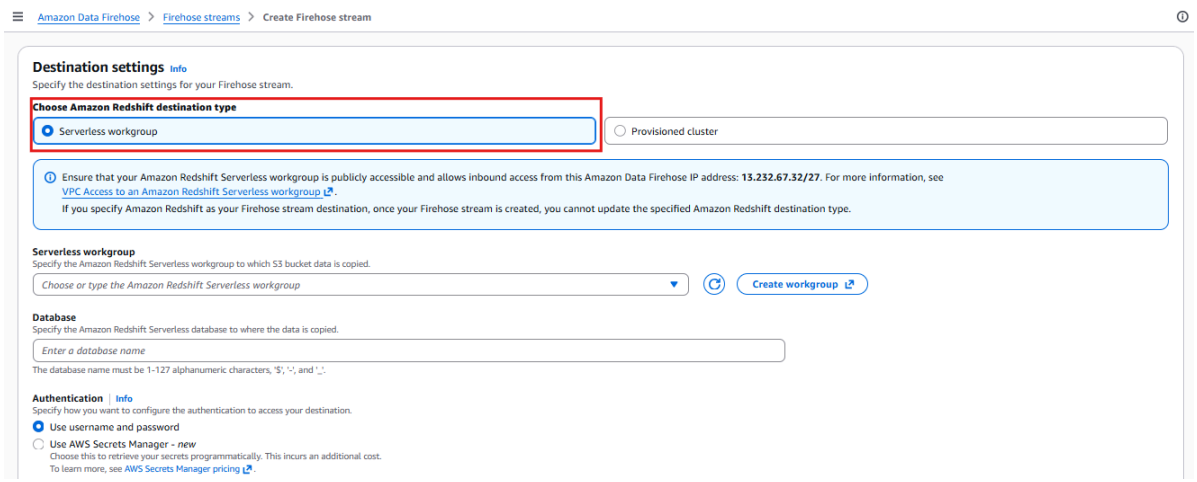
Set:

- Buffer Size → 0.2 MB
- Buffer Interval → 20 seconds

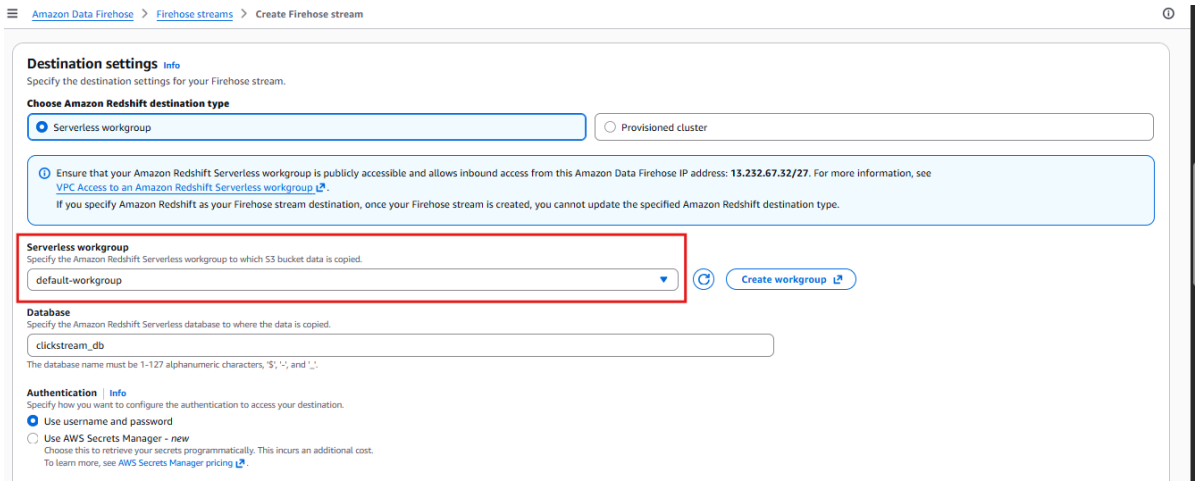


Step 4: Configure Destination (Redshift)

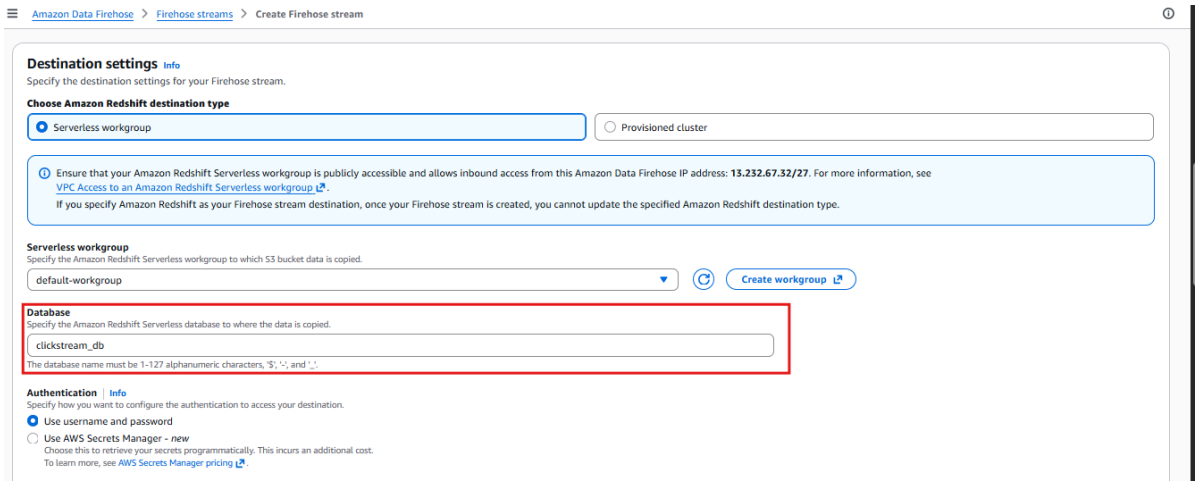
- Destination → Redshift
- **Choose Amazon Redshift destination type** —→ Serverless workgroup



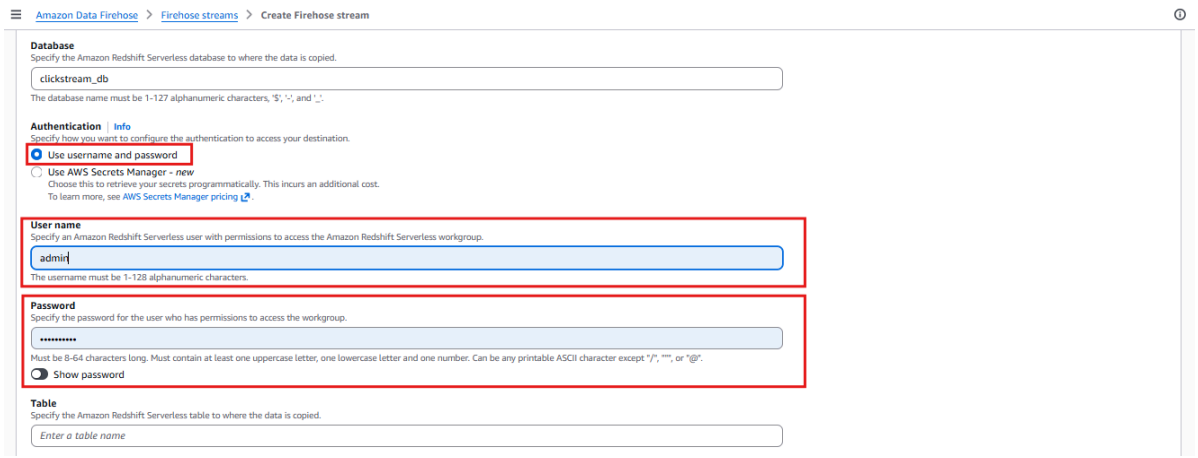
- Select Serverless workgroup



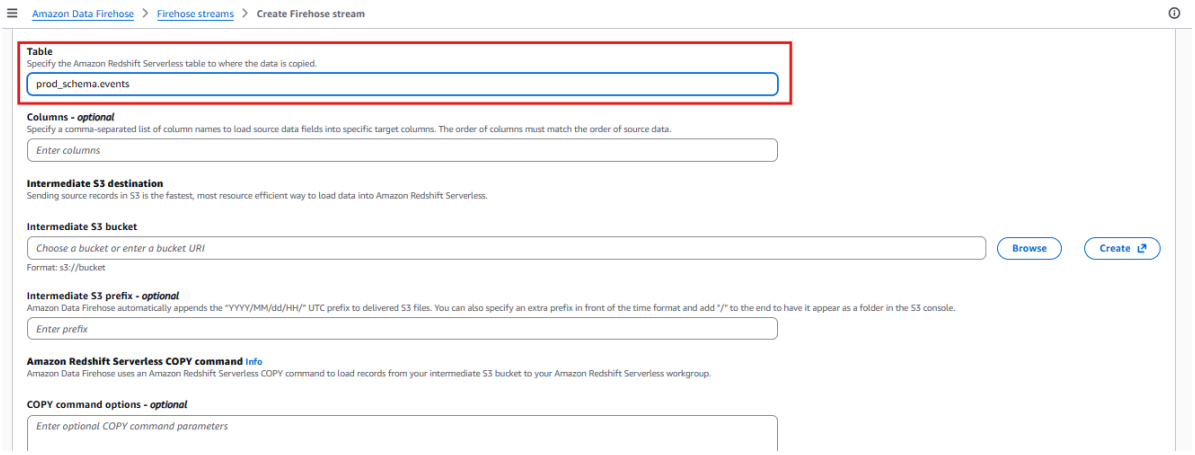
- Database → clickstream_db



- Enter Redshift username & password

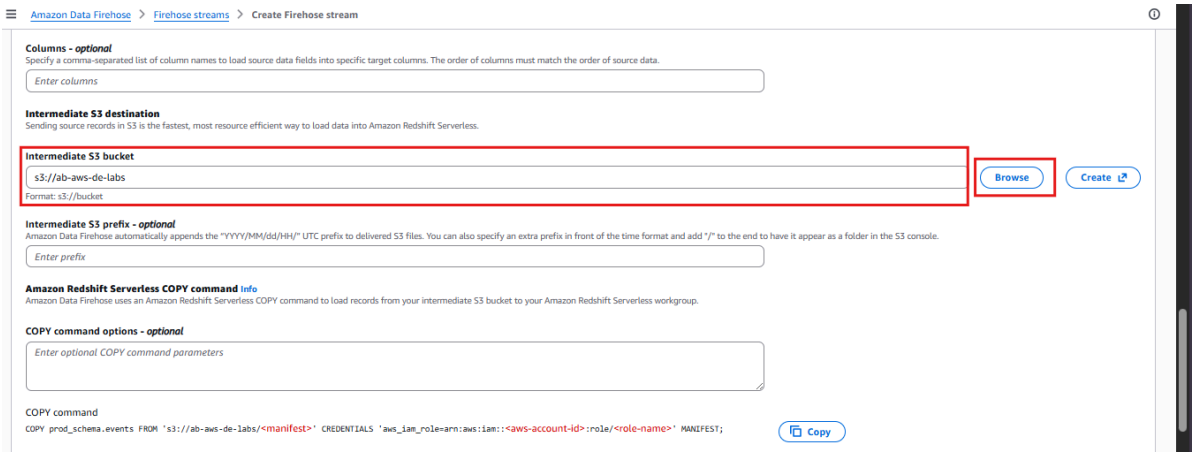


- Table → prod_schema.events



Step 5: Configure S3 Intermediate Storage

- Select S3 bucket

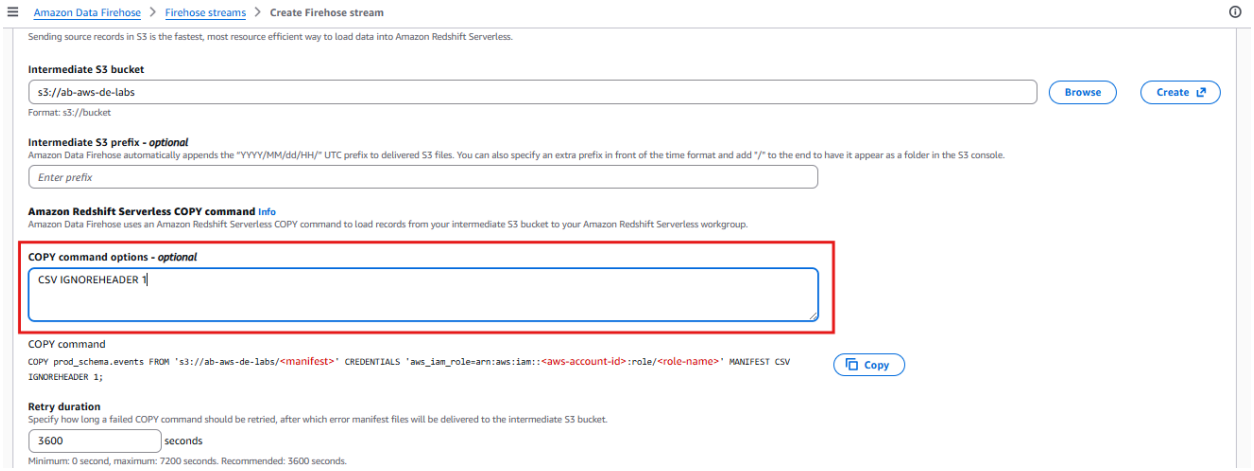


- Leave default folder structure

Step 6: Configure COPY Command

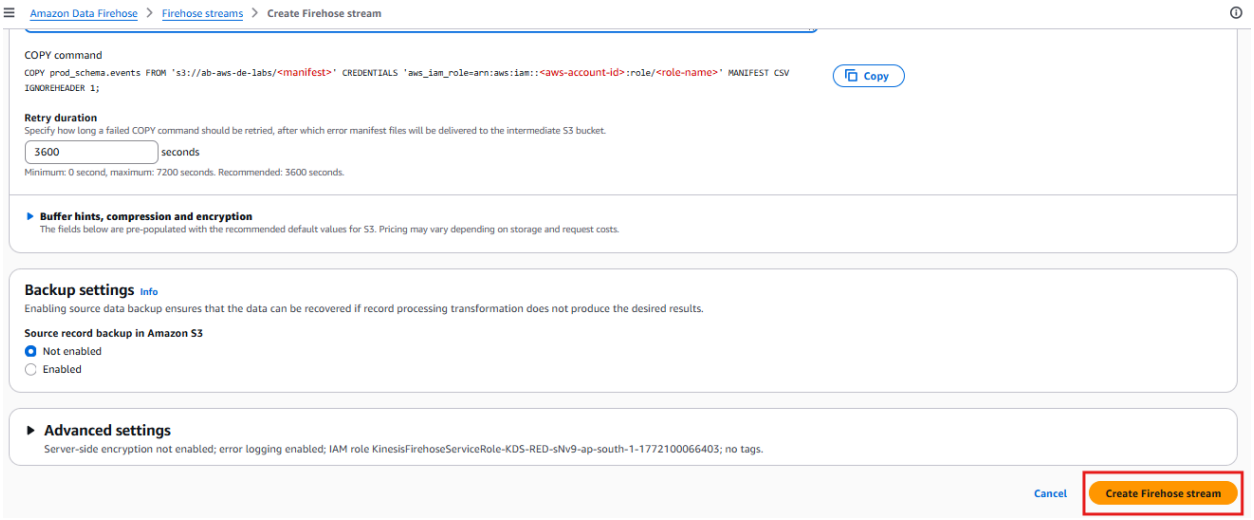
In COPY options, enter:

```
CSV IGNOREHEADER 1
```



Step 7: Create Firehose

Click → Create Firehose Stream



Task 5 — Deploy Producer Script

Step 1: Open Producer Script

Open `kinesis_events_producer.py`

Update:

```
stream_name = "app-clickstream-events"
region = "ap-south-1"
```

Step 2: Run Producer

Navigate to folder:

```
cd path/to/project
```

Run:

```
python kinesis_events_producer.py
```

You should see records being sent.

Task 6 — Monitor Pipeline

Step 1: Monitor Firehose

Go to:

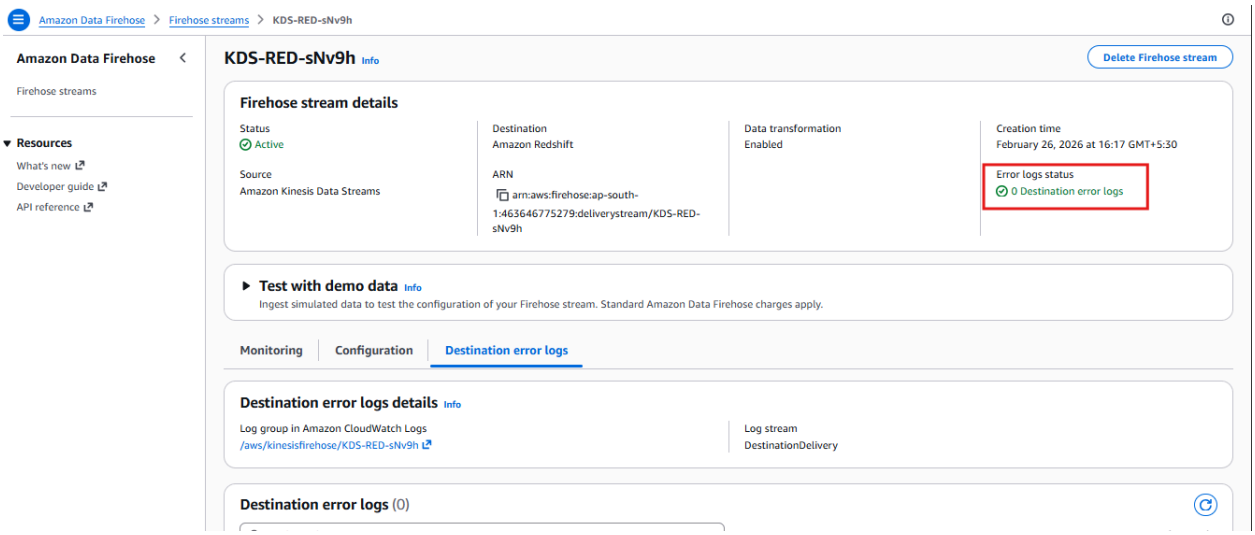
Kinesis → Firehose → Your Stream

Check:

- Incoming Records
- Delivery to Redshift
- Destination Error Logs

Ensure:

Destination Error Logs = 0

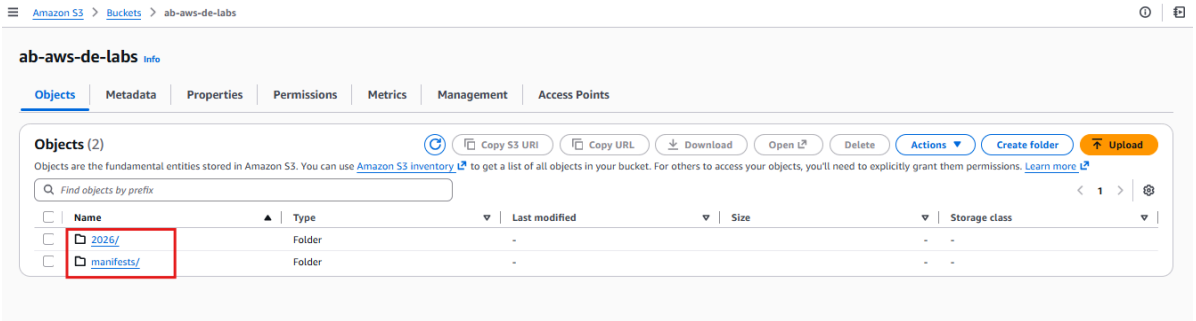


Step 2: Check S3

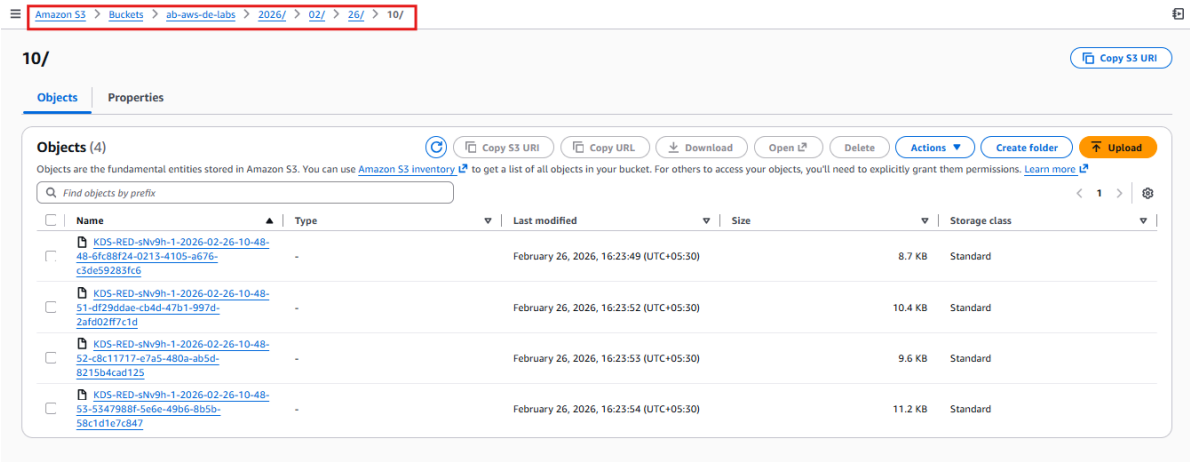
Go to S3 bucket.

You should see:

- Firehose folder structure



- CSV files without extension

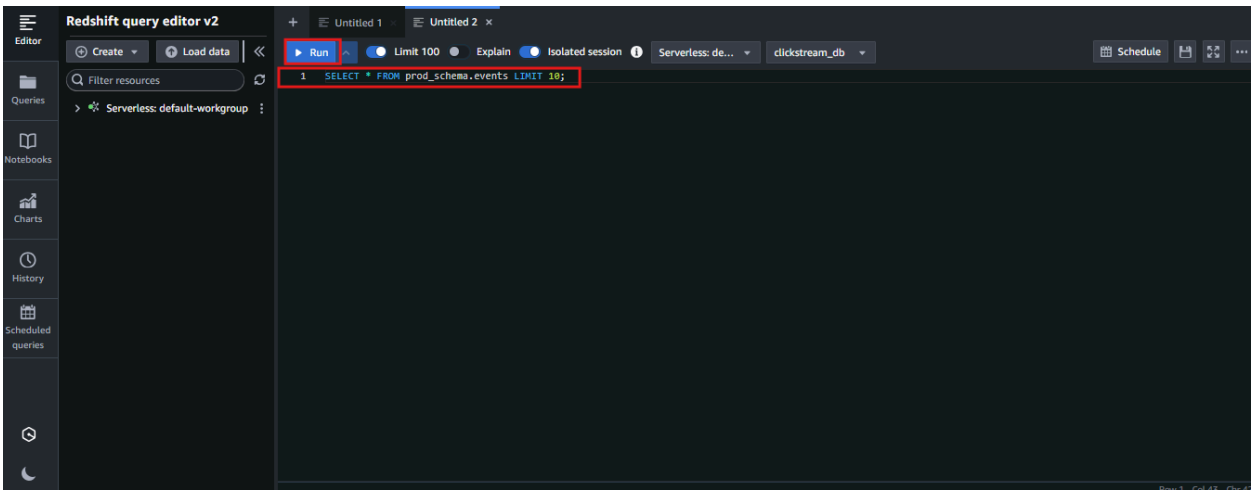


This confirms extraction phase working.

Step 3: Query Redshift

Go to Query Editor:

```
SELECT * FROM prod_schema.events LIMIT 10;
```



You should see:

- Original columns
- is_mobile column
- traffic_source_category column

Output

After completing this lab:

- Real-time data flows from Kinesis Data Stream
- Firehose buffers and invokes Lambda
- Lambda enriches records
- Firehose loads data into Redshift
- Data available for real-time reporting
- Pipeline architecture built: Kinesis → Firehose → Lambda → S3 → Redshift